

Structural Ray Tracing: A Cross-Platform Slang Module for Pipeline Ray Tracing

Kai Zhang
kazhang@nvidia.com
NVIDIA
USA

ABSTRACT

The Slang shading language compiles one shader codebase to all major GPU APIs, yet pipeline ray tracing still cannot be written once and compiled to every target: D3D12, Vulkan, and OptiX share a model of free-standing stage entry points dispatched through a host-built shader binding table (SBT), while Metal has no native *miss* or *closesthit* stages and no complete SBT record indexing. On every API, a trace call’s reachable programs are invisible to the shader compiler, leaving that boundary unchecked. We present *structural ray tracing*, an experimental Slang module in which the shader declares a *trace program schema* — the finite, typed set of hit groups, *miss* shaders, and *callable* shaders a trace can reach — while the host retains full ownership of the physical SBT. The schema buys both portability and compile-time checking: the compiler preserves native dispatch on D3D12, Vulkan, and OptiX while synthesizing Metal’s missing dispatch machinery, and on every target it checks each trace call against its reachable programs. For example, passing a payload type none of them expects fails at compile time; on today’s native APIs, the same mistake surfaces only when the GPU misreads the payload at runtime. A Cornell-box demo runs one Slang shader set on all of these APIs with no measurable GPU dispatch overhead, and a port of a Falcor 2-based renderer validates the design at mid-size-renderer scale.

CCS CONCEPTS

• Computing methodologies → Ray tracing; • Software and its engineering → Compilers.

KEYWORDS

ray tracing, shading languages, shader binding table, Slang, Metal, Vulkan, DXR, OptiX, cross-platform GPU programming

1 PROBLEM AND MOTIVATION

Every native pipeline ray-tracing API couples three parties through untyped conventions. The shader declares free-standing stage entry points; the host builds an SBT whose records name those entry points and carry per-record data; and a trace call selects records at runtime. On D3D12, Vulkan, and OptiX the driver or hardware selects each hit-group record through the same index calculation:

$$\underbrace{\text{recordIndex}}_{\text{driver/HW}} = \underbrace{\text{instanceContribution}}_{\text{host, per instance}} + \underbrace{\text{geometryIndex}}_{\text{host, per geometry}} \times \underbrace{\text{sbtStride}}_{\text{trace call}} + \underbrace{\text{sbtOffset}}_{\text{trace call}}$$

A *miss* record is instead selected directly by *missIndex*, and a *callable* record by the explicit index passed to the call; the selected stages then dispatch natively.

Metal overlaps with only the traversal half of this model. During traversal it does dispatch through a function table: each *candidate hit* — a tentative intersection not yet accepted as the closest — can invoke a triangle or procedural intersection function, Metal’s *anyhit*-style filter, selected by the sum of a per-instance and a per-geometry table offset. But the trace call contributes no *sbtStride* or *sbtOffset* analogue to that index, the dispatched functions can only accept or reject their candidate, and *miss*, *closesthit*, and *callable* have no stages at all: *miss* and *closesthit* logic is ordinary control flow after *intersect()* returns, and a *callable* is an explicit function call the shader makes itself. Three concrete gaps remain:

- **Dispatch-model gap.** No native dispatch reaches *miss* or *closesthit* code, and the traversal table index is not the full SBT record calculation: the trace-to-program mapping lives only in host pipeline setup.
- **Resource mismatch.** Metal’s function tables are shader-visible resources that must be declared and bound; the D3D12 and Vulkan SBT has no shader-side binding point at all.
- **Tag-inference gap.** Every reachable intersection function must carry a compatible `[[intersection(... )]]` tag list¹ — the static signature of the traversal capabilities the function relies on and the inputs it may consume — but reachability is host knowledge, so the compiler has no basis to infer those tag lists.

The practical consequence: engines either write and maintain their pipeline ray-tracing shaders per platform, or skip Metal entirely. And on every API, a trace call’s reachable programs are invisible to the shader compiler, so mistakes at that boundary are caught late — if ever.

2 TECHNICAL APPROACH AND KEY CONTRIBUTIONS

The design rests on one ownership split. The **shader owns the schema**: a trace program schema declares the finite set of executable hit groups, *miss* shaders, and *callable* shaders, plus the types of their payloads, hit attributes, and per-record data. The **host owns the SBT instance**: record count, order, program layout, and record bytes remain runtime host decisions, exactly as in the native APIs. Stage logic is written as stateless structs implementing stage interfaces rather than free-standing entry points (Listing 1).

¹Metal Shading Language Specification, version 4.1, §2.17.1 (Ray-Tracing Intersection Tags), §5.1.6 (Intersection Functions), and §2.17.6 (Intersector Type): “Intersection tags on the intersector type must match their associated intersection function ... or the behavior is undefined.”

```

import slang.raytracing; // experimental module

struct RadianceHitContext : rt::IHitContext
{
    typealias TraceContext = SceneTraceContext;
    typealias Payload      = RadiancePayload;
    typealias Record       = MaterialRecord; // SBT record data
    typealias Primitive    = rt::TrianglePrimitive;
}

struct RadianceClosestHit : rt::IClosestHitShader
{
    typealias Context = RadianceHitContext;
    void invoke(rt::ClosestHitInput<Context> input)
    {
        input.payload.radiance = shade(input.record,
            input.triangle.barycentricCoord);
    }
}

// The schema declares which programs exist -- not SBT records.
struct SceneSchema : rt::ITraceProgramSchema
{
    typealias TraceContext = SceneTraceContext;
    typealias HitGroups =
        rt::HitGroupList<RadianceGroup, ShadowGroup>;
    typealias MissShaders =
        rt::MissShaderList<SkyMiss, ShadowMiss>;
    typealias CallableShaders = rt::NoCallableShaders;
}

// One binding point for the whole trace program: on D3D12,
// Vulkan, and OptiX it lowers to no shader-visible resources;
// on Metal, to its function tables and record buffer.
rt::AccelerationStructure scene;
rt::TraceProgramDescriptor<SceneSchema> program;

// The payload type selects the matching programs in the schema.
rt::RayTracer<SceneSchema> tracer;
RadiancePayload p = {};
tracer.trace(desc, scene, program, p);

```

Listing 1: The new API surface. A context fixes the payload, record, and primitive types; the schema lists which programs exist; one typed descriptor binds the whole trace program; the trace call is checked against the schema.

Schema entries are code, not records. A schema entry identifies executable code — it never claims a physical SBT record, and no record index is assigned in shader source. The host still decides record count, record order, and which program each record runs, writing each record’s native shader identifier (or, on Metal, its reflected function index) itself: one entry may back records 1, 4, and 10,000, each with different record data.

Trace calls are no longer opaque. Natively, the programs a trace call may invoke are recorded only in host pipeline setup — invisible to the shader compiler. The schema gives each trace call a deliberately weak, typed connection to its reachable hit-group, *miss*, and *callable* programs: weak enough that the host still owns record placement, strong enough that mismatched payload and hit-attribute types are diagnosed at compile time rather than at pipeline creation or runtime, and that the reachable set stays visible while maintaining and debugging a large ray-tracing codebase.

One typed binding point. The shader binds the whole trace program as a single `TraceProgramDescriptor<Schema>`, closing the resource mismatch: on D3D12, Vulkan, and OptiX the descriptor lowers to no shader-visible resources — the host builds the native SBT exactly as today — while on Metal the same declaration lowers to the intersection-function tables, visible-function tables, and record buffer that Metal requires, bound through reflection.

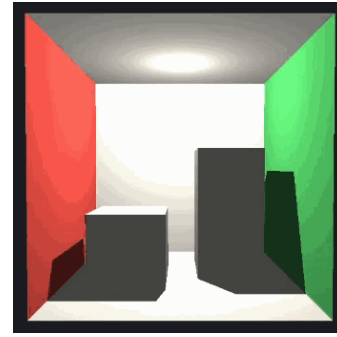


Figure 1: Cornell box rendered through structural ray tracing: one ~240-line Slang shader set compiles unmodified to SPIR-V (Vulkan), DXIL (D3D12), OptiX/CUDA, and MSL (Metal); headless renders are byte-for-byte identical (SHA-256) across the Vulkan, D3D12, and Metal runs, with matching checksums on OptiX.

Native where native exists, synthesized where it does not. On D3D12, Vulkan, and OptiX, stage structs lower to ordinary native entry points and native SBT dispatch. On Metal, the same schema drives synthesis of per-payload intersection-function-table dispatchers, visible-function tables, a compiler-owned record buffer whose lookup evaluates the standard hit-record index formula at runtime, and fully inferred `[[intersection(...)]]` tag lists.

Separate compilation. A schema section may be left open, to be completed at Slang link time by every linked type that implements a designated tag interface — so independently compiled modules (e.g., material plugins) contribute hit groups before final linking.

A reflection contract. Hosts read exact target entry symbols, Metal function indices, record layouts, and native ABI sizes from reflection — a single source of truth for building each target’s pipeline and SBT.

3 RESULTS AND DEMONSTRATIONS

A standalone Cornell-box demo (Figure 1) compiles one ~240-line Slang shader set unmodified to Vulkan and OptiX (Linux), D3D12 (Windows), and Metal (a direct Metal-cpp host consuming generated MSL on macOS). Median GPU dispatch time is identical to Slang’s existing entry-point path — the code path engines ship today — on Vulkan (0.013 ms) and D3D12 (0.035 ms), and matches a hand-written native Metal intersector within measurement noise on Apple M4 (0.088 ms vs. 0.090 ms); OptiX is validated functionally but was not perf-measured. The cost is at compile time: roughly 44–47% more compile wall time.

The in-progress compiler implementation (~52K lines across 325 files) passes 401 dedicated compiler and runtime tests, and its runtime suites pass on all of these backends — including 16 native Metal scenarios covering curves, motion, and multi-level acceleration structures. A port of a Falcor 2–based renderer exercises the design at mid-size-renderer scale: four ray-tracing consumers — including a reference path tracer whose schema serves two payload types — produce byte-identical output against the existing path

on Vulkan, pass functional validation on both D3D12 and Vulkan, and complete separate-compilation, reflection, and linking checks on Metal.

4 LIMITATIONS AND WHAT ATTENDEES WILL LEARN

This is work in progress. The design proposal is an open draft awaiting review, and the implementation is not yet merged;² the feature ships as an explicitly imported experimental module. Version one covers pipeline ray tracing only — inline ray queries are already portable — and excludes Shader Execution Reordering. The Falcor 2

Metal runtime path awaits pipeline ray-tracing support in Slang-RHI, and a port of Unreal Engine 5 is planned. The compiler cannot validate host-written SBT contents — the same trust boundary as the native APIs, made explicit through reflection.

Attendees will learn why Metal breaks the SBT dispatch model that DXR, Vulkan, and OptiX share; how declaring a typed schema in shader source lets a compiler preserve native dispatch on three targets while synthesizing it on Metal; what the compiler's trace checking can and cannot guarantee; and how a reflection-driven host workflow replaces per-platform SBT setup.

²Proposal: <https://github.com/shader-slang/spec/pull/59>. Implementation: <https://github.com/shader-slang/slang/pull/12691>. Cross-platform demo: <https://github.com/kaizhangNV/structural-rt-cornell-demo>.